# Auschwitz I
# Educational 3D Reconstruction

By

Raufun Ahsan
Roll: 2007030
Session: 2023-2024
Department of  Computer Science and Engineering
Khulna University of Engineering & Technology

# Instructed by

Dr. Sk. Md. Masudul Ahsan
Professor
Department of Computer Science and Engineering
Khulna University of Engineering & Technology

Md. Tajmilur Rahman
Lecturer
Department of Computer Science and Engineering
Khulna University of Engineering & Technology

# Agenda


- Objectives 3
- Introduction 4
- Architectural Decisions 5
- Tools Used 6
- Modelling and Geometry 7
- Lighting and Shading 8
- Feature – Day/Night Cycle 9
- Feature – Procedural Geometry 10
- Feature – Multiple Interior Scenes 11
- Feature – 2D Texture Mapping and Blending 12
- Feature – Mult Source Lighting 13
- Feature – Interactive Elements 14
- Demo Video 15

# Objectives

- Build a real-time 3D reconstruction using modern OpenGL.
- Keep architecture modular so new zones/features can be added safely.
- Demonstrate advanced graphics concepts: 3D modelling, dynamic lighting, procedural geometry, and runtime interaction.
- Deliver an interactive educational walkthrough with day-night transitions

# Introduction

## The Goal

- Auschwitz I digital reconstruction.
- Exterior + interior + dynamic systems.
- Accuracy, clarity, and interactivity.

## Key Features

- Dynamic day-night cycle
- Interactive elements and models
- Animated scenes and presets
- Textures and procedurally generated surfaces
- Multi-point views
- Free-moving camera

Fig 1: Birds-eye view of camp

# Architectural decisions

## What makes this project unique?

- Central Scene orchestrator.
- Zone-based module split.
- Primitive-driven modeling.
- Easy feature integration.

Fig 2: Main camp during day

# Tools Used

**Rendering API:** OpenGL 3.3 Core

**Language standard:** C++17

**Window/input:** GLFW

**GL loader:** GLAD

**Math:** GLM

**Image loading:** STB Image

# 3D Modelling and Geometry

## Primitives used

- Cuboids
- Spheres
- Cylinders
- Cones
- Pyramids

## Surface Geometry

- Bezier curves
- Spline curves
- Ruled surfaces

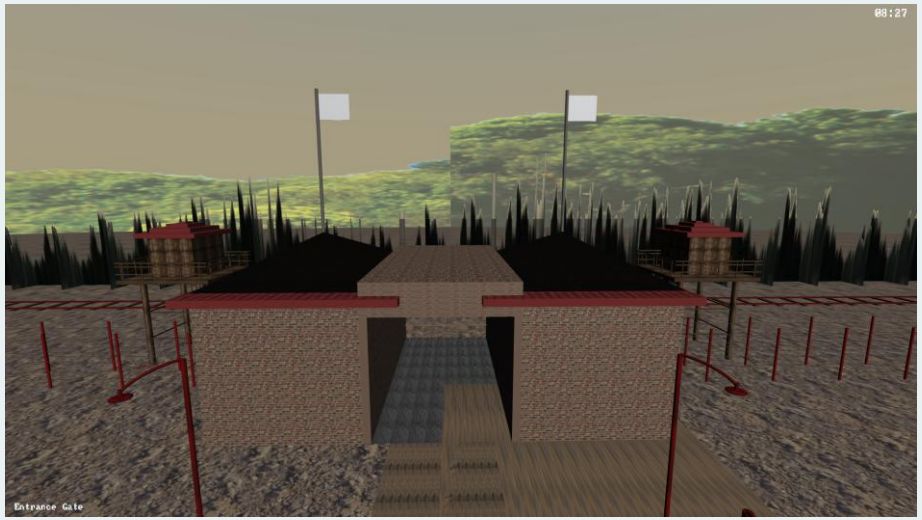



Fig 3: Main camp entrance with complex geometry

# Lighting and shading

## Lighting features

- Multi-source good looking light interaction
- Ambient, Specular and Diffuse Lighting
- Spot Lights and Point Lights

Fig 4: Daytime sky with sun

## Shading Features

- Phong Shader for clear and snappy lighting
- Multiple shader passes for diverse light effects
- Particles and celestial bodies

Fig 5: Nighttime sky with textured moon and celestial bodies

# Feature – Day/Night Cycle

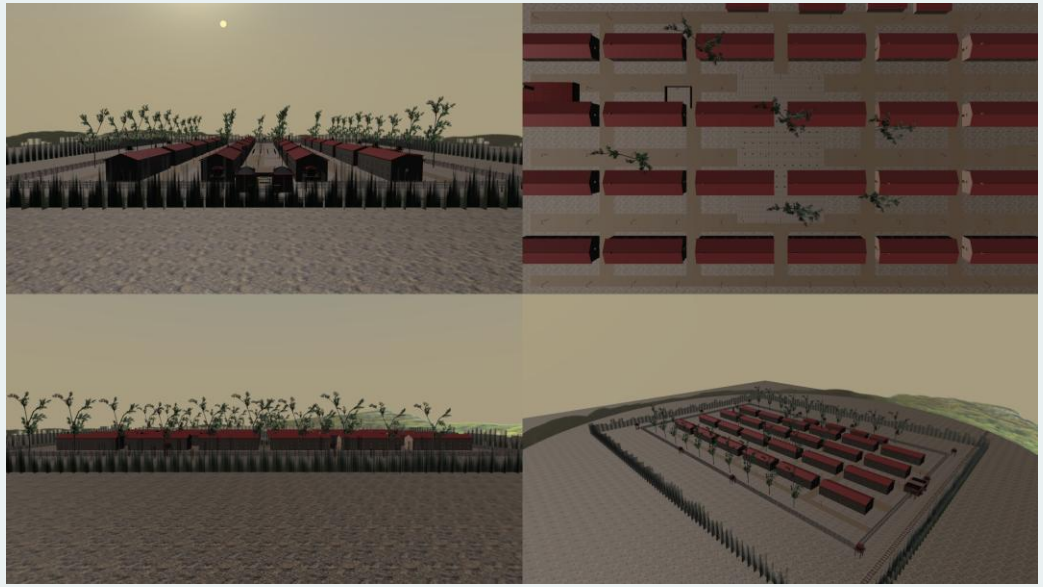


Fig 6: Camp During Day

Fig 7: Camp During Night

# Feature – Procedural Geometry

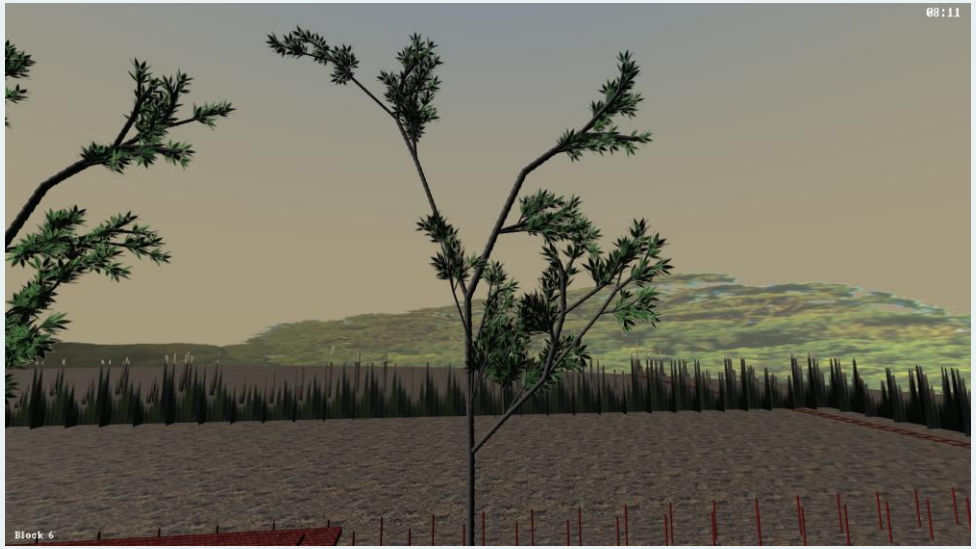



Fig 8: Tree branches generated with fractal and textured leaves

Fig 9: Coch-Fractal generated snowflakes

# Feature – Multiple Interior Scenes

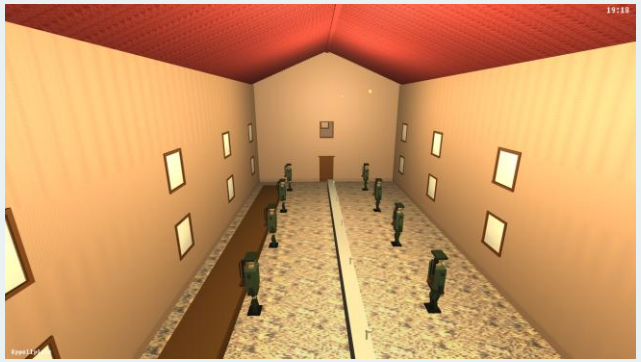


Fig 10: Interior - Army administration

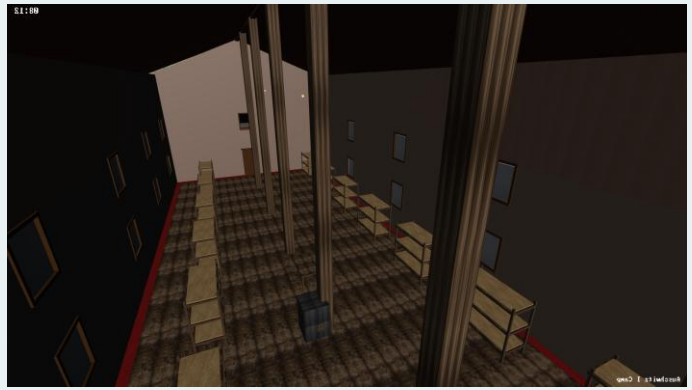


Fig 11: Interior - Living quarters

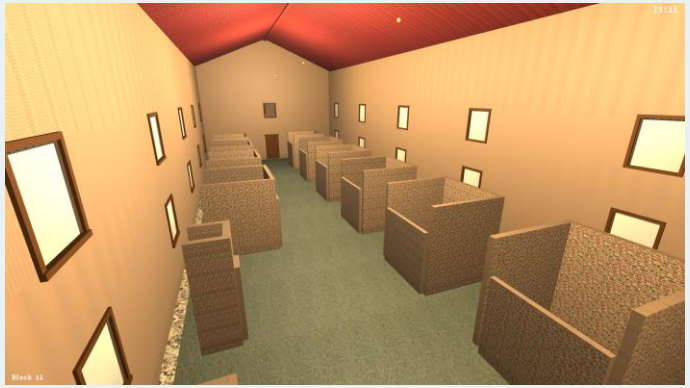


Fig 12: Interior: Jail Cells

# Feature – 2D Texture mapping and Blending

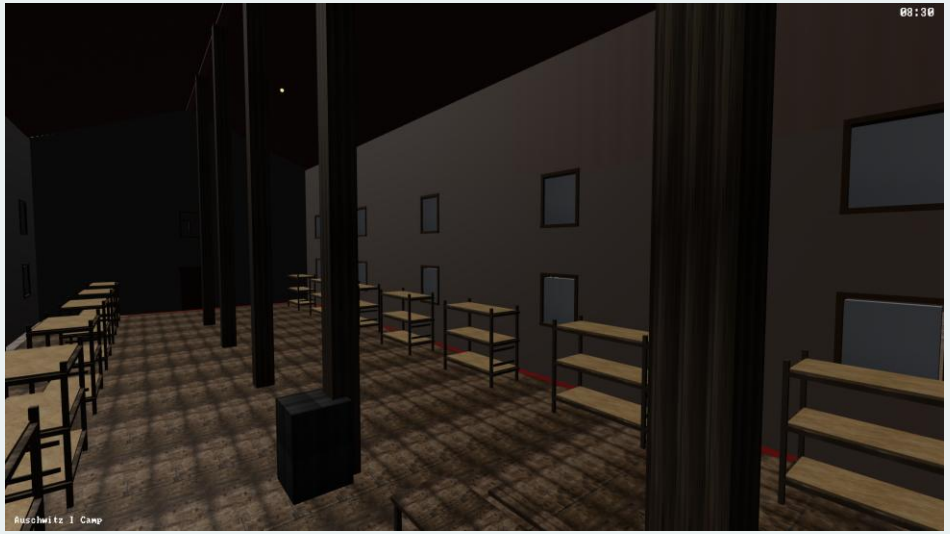



Fig 13: Textures enabled

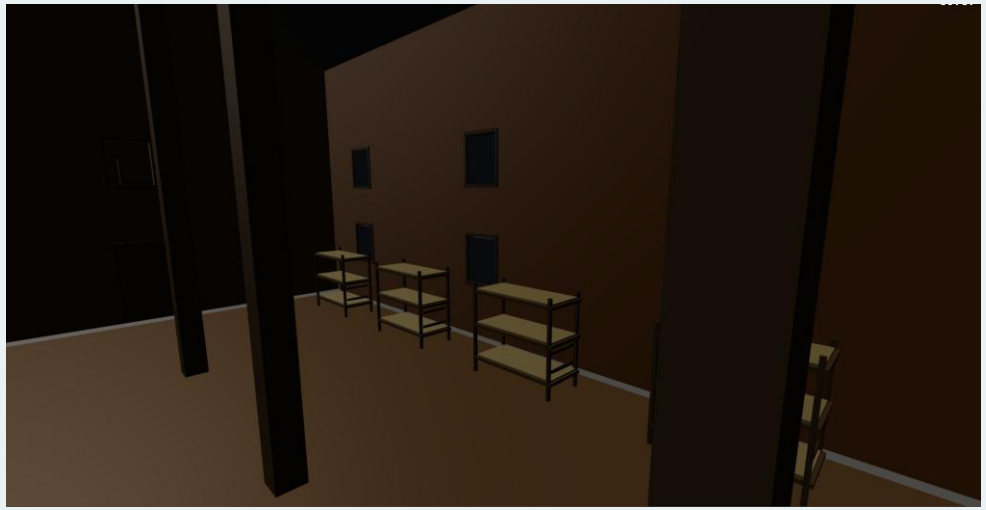


Fig 14: Textures Disabled

# Feature – Multi Source Lighting

Fig 15: Different types of lighting

# Feature – Interactive Elements

Fig 16:Interactive Army Parade

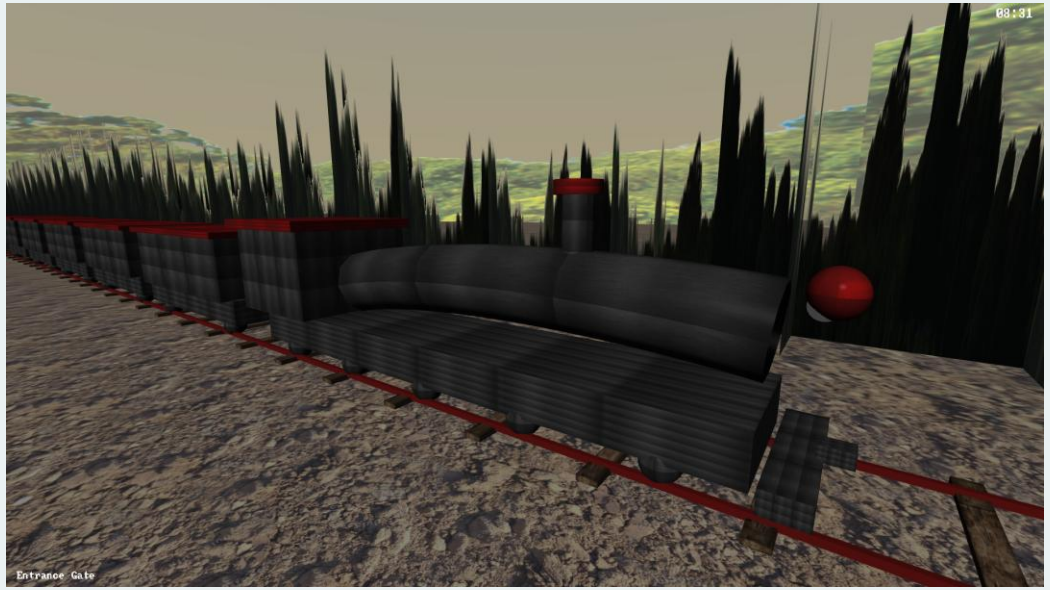
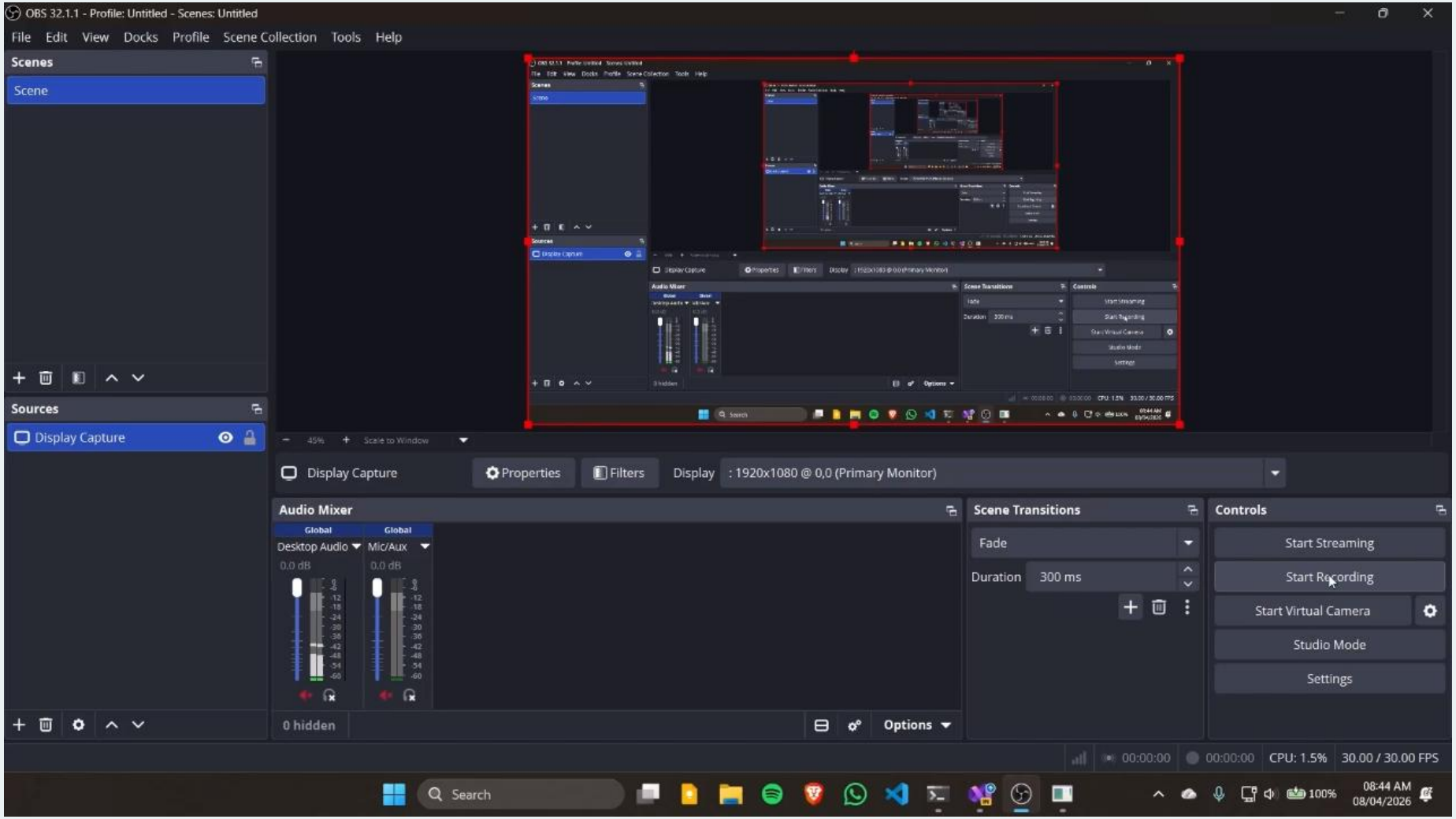



Fig 17: Departable train with multiple bogeys and engine

# Demo Video


OBS 32.1.1 - Profile: Untitled - Scenes: Untitled
File Edit View Docks Profile Scene Collection Tools Help
Scenes
Scene
Sources
Display Capture
Display Capture
Properties
Filters
Display
: 1920x1080 @ 0,0 (Primary Monitor)
Audio Mixer
Global
Desktop Audio
Mic/Aux
0.0 dB
0 hidden
Options
Scene Transitions
Fade
Duration
300 ms
Controls
Start Streaming
Start Recording
Start Virtual Camera
Studio Mode
Settings
00:00:00
CPU: 1.5%
30.00 / 30.00 FPS
Search
08:44 AM
08/04/2026

# Thank you